

Example

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1 Quarto Example

This is an example of a Quarto document, you will need to install Quarto from <https://quarto.org/>. This package comes with several common R packages, it is also recommended to install:

```
install.packages(c('gt', # general table formatting
  'gtsummary', # for univariate and regression tables
  'here', # where you are working (better when working with git)
  'knitr', # r package for working with documents
  'quarto', # R package for working with quarto
  'rmarkdown', # creating documents
  'tidymodels', # modern interface for fitting models
  'tidyverse', # modern R language
  'tinytex', # needed for pdf documents
  'devtools' # package creation development tools
))

remotes::install_github("Pakillo/grateful") # for package citation
```

To create pdf documents it is also recommended to run

```
tinytex::install_tinytex()
```

Table 1: first 5 records and 5 columns of the data

arl_number	head_shape	orientation	nom_velocity_m_sec	init_strike_velocity_m_s
SYN-001	Round	Frontal	4.67	5.71
SYN-002	Round	Lateral	5.43	4.76
SYN-003	Round	Lateral	6.00	5.20
SYN-004	Round	Lateral	4.74	6.21
SYN-005	Oval	Lateral	5.07	6.50

2 Table One

3 Cross Reference

You can cross-reference tables, listings, figures and page numbers

You can use the `labels` you created to reference the tables like Table 2 or items in the `bib.bib` file such as Al-Helali et al. (2021).

You can even reference later items such as Figure 2 or sections like Section 5 or even the page number that something happens on like page 7.

4 Pairwise t-tests

Suppose we wanted to visualize all pairwise t-tests (we will only do first four but you get the idea). There are other ways to do this but this is a good enough example of repeated analysis.

Table 2: An example of a Table 1

Characteristic	Oval N = 33 ¹	Round N = 27 ¹	p-value	Overall ¹
orientation			0.94 ²	
Frontal	13 (39%)	12 (44%)		25 (42%)
Lateral	14 (42%)	10 (37%)		24 (40%)
Occipital	6 (18%)	5 (19%)		11 (18%)
nom_velocity_m_sec			0.67 ³	
Mean (SD)	5.43 (0.80)	5.34 (0.71)		5.39 (0.76)
Median (Q1, Q3)	5.40 (5.07, 5.87)	5.43 (4.67, 5.99)		5.40 (4.77, 5.92)
Median (Min - Max)	5.40 (3.88 - 7.66)	5.43 (4.32 - 6.71)		5.40 (3.88 - 7.66)
init_strike_velocity_m_s			0.62 ³	
Mean (SD)	5.19 (0.80)	5.12 (0.71)		5.16 (0.76)
Median (Q1, Q3)	5.17 (4.77, 5.70)	4.99 (4.72, 5.73)		5.14 (4.74, 5.72)
Median (Min - Max)	5.17 (3.63 - 6.72)	4.99 (3.76 - 6.36)		5.14 (3.63 - 6.72)
depth_mm			0.27 ³	
Mean (SD)	3.60 (2.00)	3.99 (1.63)		3.77 (1.83)
Median (Q1, Q3)	3.34 (1.83, 4.68)	3.72 (2.88, 5.19)		3.51 (2.52, 4.92)
Median (Min - Max)	3.34 (0.76 - 8.41)	3.72 (1.39 - 7.66)		3.51 (0.76 - 8.41)
max_depth_mm			0.29 ³	
Mean (SD)	5.35 (2.25)	5.92 (2.12)		5.61 (2.19)
Median (Q1, Q3)	4.95 (3.70, 6.94)	5.45 (4.30, 7.49)		5.31 (3.90, 7.10)
Median (Min - Max)	4.95 (1.68 - 10.40)	5.45 (2.83 - 10.69)		5.31 (1.68 - 10.69)
duration_ms			0.29 ³	
Mean (SD)	6.2 (2.7)	7.2 (3.7)		6.7 (3.2)
Median (Q1, Q3)	6.0 (4.9, 7.6)	6.4 (4.1, 8.7)		6.1 (4.6, 7.9)
Median (Min - Max)	6.0 (2.0 - 12.9)	6.4 (1.3 - 15.8)		6.1 (1.3 - 15.8)
transf_energy_j			0.68 ³	
Mean (SD)	4.38 (2.22)	3.99 (1.89)		4.21 (2.07)
Median (Q1, Q3)	4.10 (2.59, 5.43)	3.65 (2.36, 5.19)		3.78 (2.54, 5.42)
Median (Min - Max)	4.10 (1.80 - 10.83)	3.65 (0.81 - 8.23)		3.78 (0.81 - 10.83)

¹n (%)²Fisher's exact test³Wilcoxon rank sum exact test

4.1 between nom_velocity_m_sec and init_strike_velocity_m_s

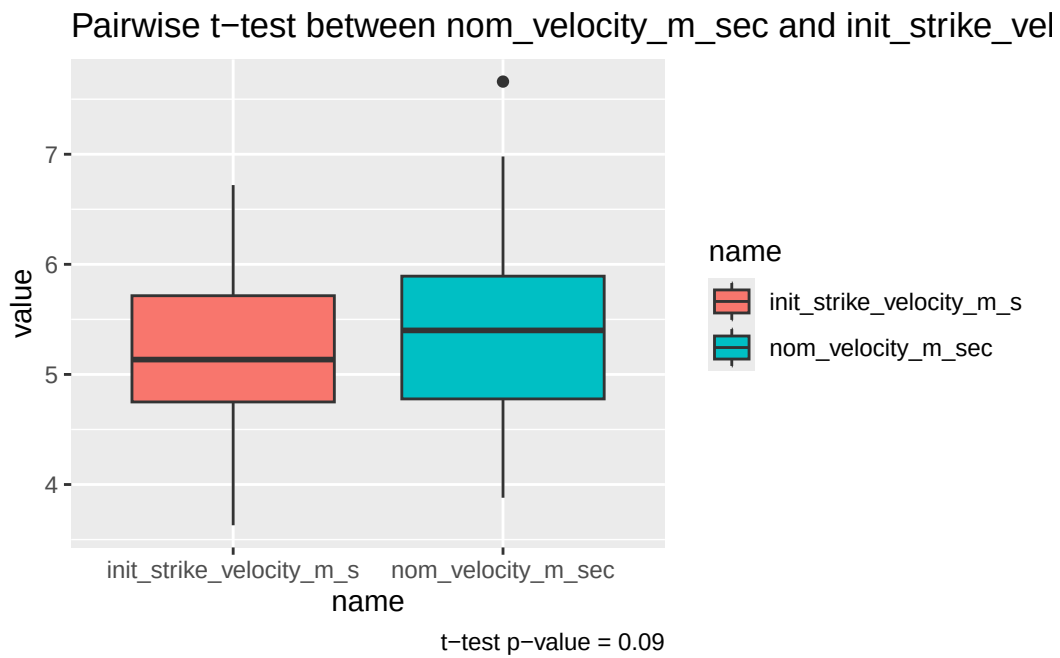


Figure 1: Pairwise t-test between nom_velocity_m_sec and init_strike_velocity_m_s

4.2 between nom_velocity_m_sec and depth_mm

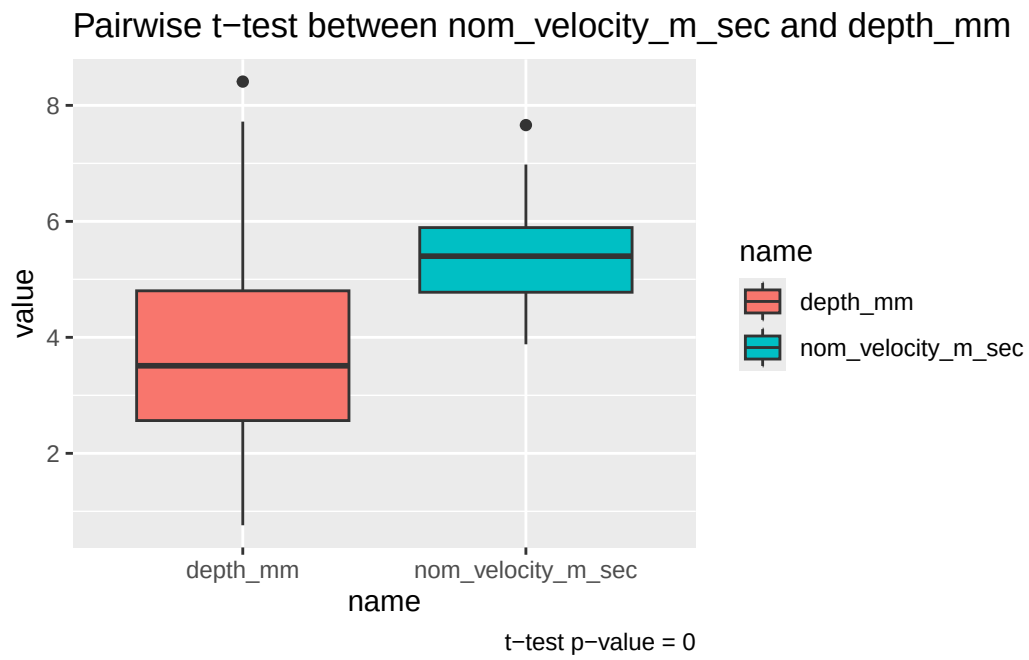


Figure 2: Pairwise t-test between nom_velocity_m_sec and depth_mm

4.3 between nom_velocity_m_sec and max_depth_mm

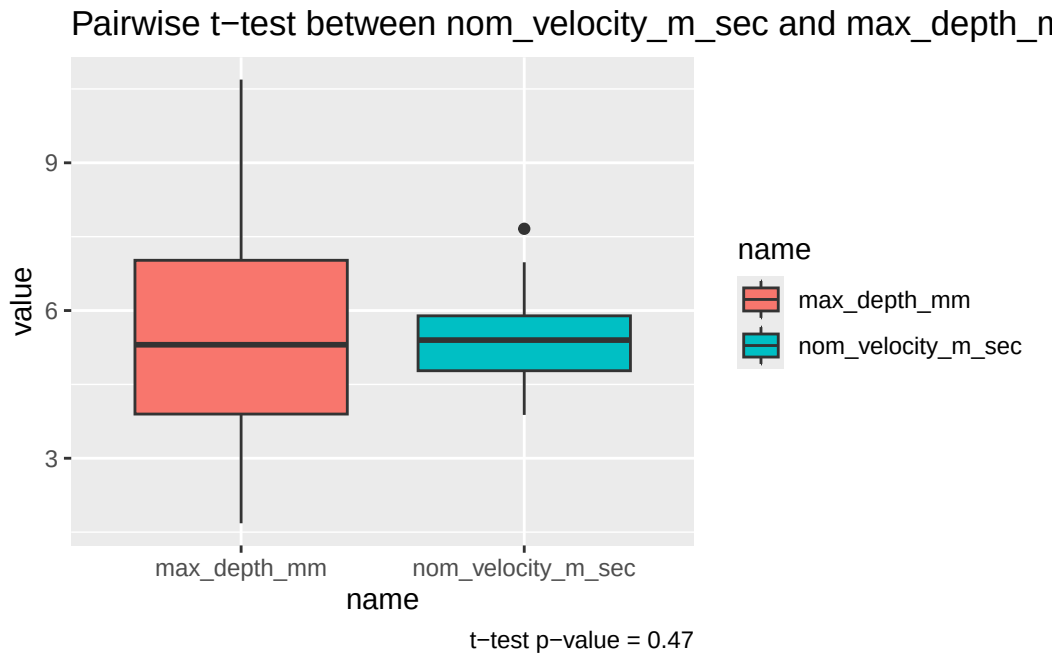


Figure 3: Pairwise t-test between nom_velocity_m_sec and max_depth_mm

4.4 between nom_velocity_m_sec and duration_ms

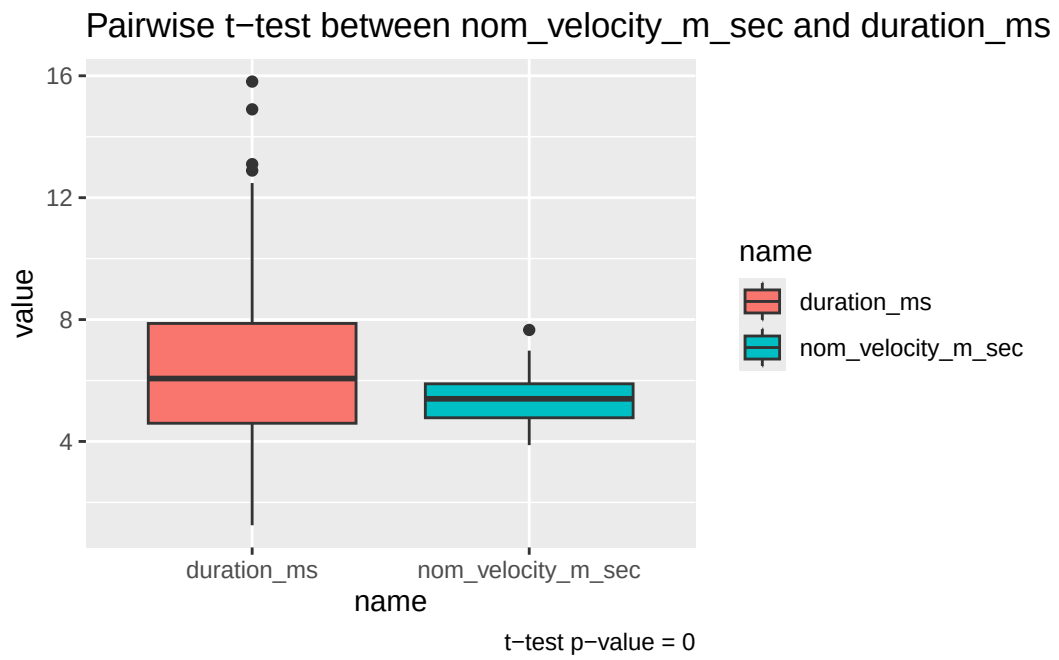


Figure 4: Pairwise t-test between nom_velocity_m_sec and duration_ms

5 Software

This work was completed using R v. 4.6.1 (R Core Team 2026) and the following R packages: bookdown v. 0.48 (Xie 2016, 2026), grateful v. 0.3.0 (Rodriguez-Sanchez and Jackson 2025), gt v. 1.3.0 (Iannone et al. 2026), gtsummary v. 2.5.1 (Sjoberg et al. 2021), here v. 1.0.2 (Müller 2025), knitr v. 1.51 (Xie 2014, 2015, 2025), rmarkdown v. 2.31 (Xie et al. 2018, 2020; Allaire et al. 2026), tidyverse v. 2.0.0 (Wickham et al. 2019).

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